LESSON 11.8
BARBIE BUNGEE – PART 2

LESSON INTRODUCTION

MA.8.DP.1.1 Given a set of real-world, bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

MA.8.DP.1.3 Given a scatter plot with a linear association, informally fit a straight line.

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and

y-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

LEARNING TARGETS

- I can collect data for an experiment.
- I can construct a scatter plot for a set of bivariate data.
- I can describe patterns of associations in data on a scatter plot.
- I can informally fit a straight line to data on a scatter plot.
- I can interpret the slope and y-intercept of a linear equation in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical, and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways, state whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.

ESSENTIAL QUESTIONS

- What kind of data can be modeled with a scatter plot?
- How does changing one test variable affect the data?
- How does the line of fit help me interpret the data from the activity?
- How is it possible to have multiple different answers that are correct?

LESSON NARRATIVE

Barbie has bungeed, and now it's time to look at the results. Students take their data from the previous lesson in 11.7.2 Activity and create a scatter plot. Students then work cooperatively in their groups to determine an appropriate line of fit and interpret the slope of the line.

COMPONENT SUMMARY

11.8.1 WARM-UP *(~5 MIN.)*

What's going on in the graph?

11.8.2 ACTIVITY *(30-35 MIN.)*

Create a scatter plot and interpret the results

11.8.3 BRING IT TOGETHER *(5 MIN.)*

Compare the line of fit to a sample student's line of fit

11.8.4 WRAP-UP *(~5 MIN.)*

Self-assess understanding and interest in the lesson

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Data from the previous lesson
- Straightedges
- (Optional) Graphing utilities for lines of fit

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may misinterpret their data for their scatter plots.
- Students may have difficulty articulating their interpretations of their data.
- Students may mislabel their axes thereby affecting their interpretations.

THINGS TO CONSIDER

- This is the second of a two-day activity where students plot and analyze their data.
- It is helpful for students to understand that not every group had the same data, yet that doesn't mean they are incorrect.

LITERACY AND LANGUAGE ROUTINES

Group Presentations

11.8.3 Bring It Together provides an opportunity for students to share their lines of fit and interpretations with the class. Consider allowing groups to present the aforementioned information with the class as a way to assist student understanding of how it's possible to have multiple correct answers.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students are provided with the opportunity to "use models and methods to understand, represent, and solve problems," as stated for MA.K12.MTR.7.1. Consider discussing with students in 11.8.3 Bring It Together how this lesson affected their understanding of scatter plots and lines of fit now that they recorded the data and made their own line of fit as opposed to being given that information to interpret.

DIFFERENTIATE INSTRUCTION

This lesson utilizes multiple representations for interpreting key features. Representations include function notation, standard form, written descriptions, table of values, and graphs. The multiplicity of representations provides a variety of entry points for learners to engage with the content.

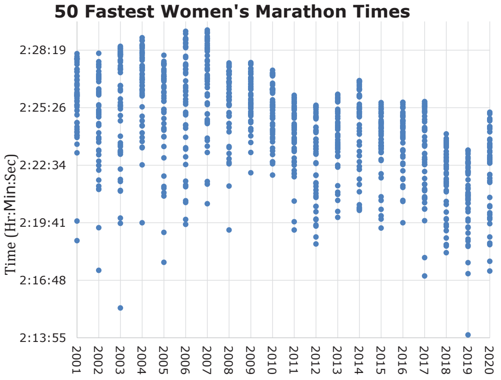
11.8.1 WARM-UP: WHAT'S GOING ON IN THE GRAPH?

TEACHER MOVES

Consider using the graph as an opportunity to have students complete a notice and wonder about the data presented. What do students notice? How do the times change yearly?

11.8.1 Warm-Up: What's Going on in the Graph?

1. The graph shows the times of the 50 fastest women's marathoners in the world from 2001 to 2020.



   a. Describe how you know if the runners are getting faster or slower each year.
   Answers will vary.
   To determine if they are getting faster or slower, I look at the overall trend of the data.

   b. Give at least one possible explanation for the trend.
   Answers will vary.
   I can tell that on average the women are getting faster because the data is trending in a negative direction which means their time is decreasing.

11.8.2 Activity: Barbie Bungee - Part 2

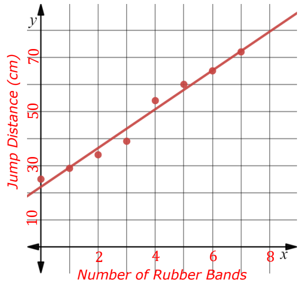
In the previous lesson, you performed an experiment and recorded the data for Barbie's bungee experience. Work with your group to represent and analyze the data collected.

Representing and Analyzing the Data

1. Construct a scatter plot to represent the data collected, using the average of the trials for each number of rubber bands.
   - Label the axes with the appropriate variables.
   - Identify and label an appropriate scale for each axis on the grid.
   - Plot a point for each pair of values in the data set.
   - Title the graph.

   Answers will vary based on student records.
   Sample scatterplot:

   Barbie Bungee Results



2. Interpret the data represented on your scatter plot.

   a. Describe the relationship between the number of rubber bands used and the jump distance.
   Answers will vary.
   As the number of rubber bands increases the jump distance increases.

   b. Describe any patterns of association in the scatter plot.
   Answers will vary.
   The scatterplot has a positive linear association.

   c. Informally fit a line to the data using a straightedge.
   Answers will vary.

3. Discuss the following questions with your partner.

   a. How do you think the data would change if Barbie were taller?

   b. How do you think the data would change if Barbie were lighter?

11.8.2 ACTIVITY: BARBIE BUNGEE – PART 2

DIFFERENTIATE INSTRUCTION

Consider other forms of representation for student scatter plots by allowing them to complete the scatter plots on the computer or enlarged graphs.

DIFFERENTIATE INSTRUCTION

Consider allowing student responses for question 2 to be presented in multiple manners, for example verbally, or using diagrams or drawings.

ENRICHMENT

Students are not expected to write their answers to question 3. Monitor student discussions to analyze student hypotheses. Look for student understanding of the activity in their discussion.

11.8.3 BRING IT TOGETHER: INTERPRETING DATA

TEACHER MOVES

Group Presentations

Consider using this section as an opportunity to implement the language routine, Group Presentations. Students can share their lines and their interpretations. They can then be asked to compare their answers to other groups to understand their interpretations of scatter plots more completely.

11.8.3 Bring it Together: Interpreting Data

1. Noah performed the same Barbie Bungee experiment. He fit the line $y = 7.2x + 22.17$ to the data in his scatter plot, where x represents the number of rubber bands used and y represents the jump distance in centimeters.

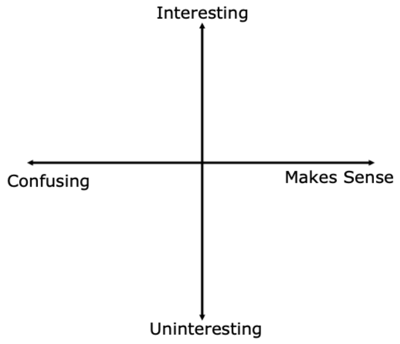
   a. Interpret the slope of the line fit to Noah's data.
   Answers will vary. On average for each additional rubber band added, barbie will travel 7.2 *cm. further.*

   b. Discuss with your partner whether or not you think Noah's line could be a good fit for the data you collected in the experiment.

   Answers will vary. I think that Noah's line could be a good fit for my data because the slope of my line was very close to the slope of the line Noah drew.

11.8.4 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your thoughts about the Barbie Bungee activity.



2. Explain why you placed your point there.
   Answers will vary.

11.8.4 WRAP-UP: REFLECTION

QUESTIONING

For students who finish early, ask them to continue their answer for question 2 by writing specific elements that they found most interesting, and constructively write elements of ways that could have made it even better.

